Maintenance Schedule

Diesel Engine

EPA/CARB
12V4000S83-E
12V4000S83LE
With BiFuel
Application Group 4D

MSE50206/01E

Power. Passion. Partnership.

Engine model	kW/cyl.	Application group
12V4000S83-E	139,8 kW/cyl.	4D, Frac-operation (continuous operation with low load)
12V4000S83LE	155,4 kW/cyl.	4D, Frac-operation (continuous operation with low load)

Table 1: Applicability

Table of Contents

DCL-ID: 0000038029 - 004

1 Maintenance Schedule

1.1 Preface

The maintenance schedule is applicable to products certified in accordance with U.S. emissions regulations and operated in the area of applicability of these regulations.

Information about the latest maintenance schedule is available at: http://www.mtu-online.com.

The following instructions must be observed for these products:

The maintenance schedule is an integral part of the emissions certification. Nonobservance of the maintenance instructions can result in violation of the statutory emission specifications. Modification or removal of mechanical or electronic components or the installation of additional components as well as the execution of calibration processes that might affect the emission characteristics of the product are prohibited by emission regulations. Emission-relevant components may only be maintained, exchanged or repaired if the components used for this purpose are approved by MTU or equivalent components. Emission-relevant components are for instance control units, data records, sensors, injectors, exhaust flaps ands all components of exhaust aftertreatment systems.

The maintenance system for MTU products is based on a preventive maintenance concept. Preventive maintenance facilitates advance operational planning and ensures a high level of equipment availability.The maintenance intervals and the scope of required maintenance activities are the result of operational experience and therefore represent recommendations. Severe operating and ambient conditions may require additional maintenance work and/or modification of the maintenance intervals.

Maintenance activities are specified with intervals based on hours in operation and time limits. Whichever value is reached first shall apply.

The individual maintenance intervals are classified in accordance with the qualification levels QL1 to QL4.

QL1: Operational monitoring and maintenance work which does not require disassembly of the product.
QL2: Exchange of components and parts (corrective only, not part of the maintenance schedule).
QL3: Maintenance work which requires partial disassembly of the product.
QL4: Maintenance work which requires complete disassembly of the product.

Additional notes on maintenance and preservation:

For change intervals for fluids and lubricants as well as for preservation specifications, refer to the relevant instructions of the component manufacturers and the MTU Fluids and Lubricants Specifications. The latest version for drive systems is available online at: http://www.mtu-online.com, and for power generation systems at: http://www.mtuonsiteenergy.com.

Components which are not listed in this maintenance schedule must be serviced in accordance with Manufacturer's Instructions.

TIM-ID: 0000046248 - 005

1.2 Allocation matrix application group 4D

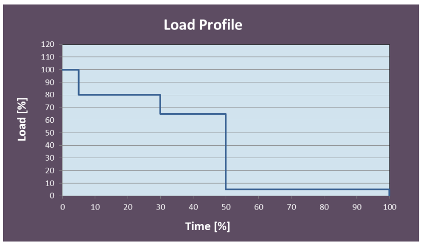
V 4000 S83 / S83 L								
Load profile		Load factor %	Load indicator %	0 - 140 kW/cyl.	141 - 156 kW/cyl.			
Load %	Time %							
110 (100-110)	–	0 - 41	≤ 9	13,500	10,000	–	–	
100 (90-100)	5							
90 (80-90)	–							
80 (65-80)	25							
65 (50-65)	20							
50 (30-50)	–							
30 (5-30)	–							
Idle (0-5)	50							

Table 2: Allocation matrix application group 4D

1.3 Maintenance schedule matrix 13,500 h

0 to 13,500 Operating hours

Task	Limit	Daily	500	1,000	1,500	2,000	2,500	3,000	3,500	4,000	4,500	5,000	5,500	6,000	6,500	7,000	7,500	8,000	8,500	9,000	9,500	10,000	10,500	11,000	11,500	12,000	12,500	13,000	13,500
Engine																													
W0800	-																												
W1024	1 w																												
W1974	1 a																												
W1008	2 a																												
W0500	-	X																											
W0501	-	X																											
W0503	-	X																											
W0505	-	X																											
W0506	-	X																											
W0525	-	X																											
W1089	-	X																											
W1009	2 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1864	6 m			X		X		X		X		X		X		X		X		X		X		X		X		X	
W1463	1 a			X		X		X		X		X		X		X		X		X		X		X		X		X	
W1001	2 a			X		X		X		X		X		X		X		X		X		X		X		X		X	
W1241	2 a			X		X		X		X		X		X		X		X		X		X		X		X		X	
W1575	1 a							X						X						X						X			
W1207	2 a			X				X						X						X						X			
W1046	2 a										X									X									X
W1011	6 a										X									X									X
W1026	6 a										X									X									X
W1526	18 a										X									X									X
W1006	18 a										X									X									X
W1713	18 a										X									X									X
W1689	18 a										X									X									X
W1610	18 a																												X
W1755	18 a							X						X						X						X			
W1250	5 a										X									X									X
W1975	18 a										X									X									X
W1251	6 a																												X
W2000	18 a																												X
W2001	18 a																												X
W2002	18 a																												X
W2004	18 a																												X
W2007	18 a																												X
W2009	18 a																												X
W2010	18 a																												X
W2062	18 a																												X
W2070	18 a																												X
W2110	18 a																												X
W2154	18 a																												X
W1051	18 a																												X

w = weeks
m = months
a = years

TIM-ID: 0000061291 - 003

Task		Operating hours [h]																													
	Limit	Daily	500	1,000	1,500	2,000	2,500	3,000	3,500	4,000	4,500	5,000	5,500	6,000	6,500	7,000	7,500	8,000	8,500	9,000	9,500	10,000	10,500	11,000	11,500	12,000	12,500	13,000	13,500		
W1043	18 a																												X		
W1134	18 a																												X		
W1605	18 a																												X		
W1735	18 a																												X		
W1041	18 a																												X		
W1035	6 a																												X		
W3000	18 a																												X		
W3001	18 a																												X		
W3002	18 a																												X		
W3003	18 a																												X		
W3004	18 a																												X		
W3005	18 a																												X		
W3007	18 a																												X		
W3018	18 a																												X		
W3021	18 a																												X		
W3083	18 a																												X		
W3097	18 a																												X		
W3100	18 a																												X		
W3103	18 a																												X		
W3108	18 a																												X		
W3112	18 a																												X		
W3143	18 a																												X		
W3177	18 a																												X		
W3182	18 a																												X		
W1124	18 a																												X		
W1331	18 a																												X		
X5374	18 a																												X		

w = weeks
m = months
a = years

TIM-ID: 0000061291 - 003

1.4 Maintenance tasks 13,500 h

Qualification level	Interval [h]	Limit	Item	Maintenance tasks	Option	Task
Engine						
QL 1	-	-	EPA label	Check if emission label is fitted and legible, and if it contains the correct information.		W0800
QL 1	-	1 w	Emergency air shut-off flaps	Check operation of emergency air shut-off flaps (electrical).		W1024
QL 1	-	1 a	Gas train	Check valves for leaks.		W1974
QL 1	-	2 a	Engine oil filter	Fit new engine oil filters each time the engine oil is changed or, at the latest, on expiry of the time limit (given in years).		W1008
QL 1	Daily	-	Engine operation	Check engine oil level.		W0500
				Carry out visual inspection of engine for general condition and leaks.		W0501
				Inspect service indicator of air filter.		W0503
				Check relief bores of coolant pump(s).		W0505
				Check for abnormal running noises, exhaust gas color and vibration.		W0506
				Inspect battery-charging generator for contamination and clean if necessary.		W0525
QL 1	Daily	-	Gas feed	Inspect lines carrying gas for leaks. Seal if necessary.		W1089
QL 1	500	2 a	Centrifugal oil filter	Check thickness of oil residue layer. Clean. Fit new sleeve, at the latest, each time the engine oil is changed.	X	W1009
QL 1	1000	6 m	Automatic engine oil filter	Check and clean oil indicator filter.	X	W1864
QL 1	1000	1 a	Engine mounts	Carry out visual inspection of engine mounts for general condition.	X	W1463
QL 1	1000	2 a	Fuel filter	Fit new fuel filter or new fuel filter insert.		W1001
QL 1	1000	2 a	Belt drive	Inspect condition of drive belts and fit new ones if necessary. Adjust tension.		W1241
QL 1	3000	1 a	Engine mounts	Carry out visual inspection of Trunnion mounts for general condition.		W1575
QL 1	3000	2 a	Valve gear	Check valve clearance, adjust if required. ATTENTION! Initial adjustment after 1,000 operating hours and subsequently 1,000 operating hours after each cylinder-head overhaul..		W1207
QL 1	4500	2 a	Crankcase breathers	Fit new filters or filter inserts.		W1046
QL 1	4500	6 a	Combustion chambers	Inspect cylinder chambers using endoscope.		W1011
QL 1	4500	6 a	Engine mounts	Check buffer clearance of mounts, check proper seating of securing screws.	X	W1026
QL 1	4500	18 a	Oxygen sensor	Fit new sensor.		W1526
QL 1	4500	18 a	Fuel injectors	Replace fuel injectors.		W1006
QL 1	4500	18 a	Engine governor	Injector: reset drift compensation parameters (CDC).		W1713
QL 1	4500	18 a	Cylinder heads	Measure valve stem end to cylinder head top distance.		W1689
QL 1	13500	18 a	Engine mounts	Check securing screws of trunnion mounts for secure seating.		W1610
QL3	3000	18 a	Flame arrester	Check for contamination, clean if necessary.		W1755
QL3	4500	5 a	Rubber sleeves	Replace all rubber sleeves.		W1250
QL3	4500	18 a	Diesel oxidation catalyst	Replace inserts.		W1975
QL3	13500	6 a	Hose lines	Replace all hose lines.		W1251
QL3	13500	18 a	Component maintenance	Before starting maintenance work, drain the coolant and flush the cooling systems.		W2000
				Inspect rocker arms and valve bridge for wear. Insert an endoscope through the pushrod bore to visually inspect swing followers and camshaft running surfaces.		W2001
				Clean air ducting.		W2002
				Fit new high-pressure fuel sensor.		W2004
				Check charge-air coolant thermostat and fit new thermal actuator.		W2007
				Inspect centrifugal oil filter for wear.	X	W2009

w = weeks
m = months
a = years

TIM-ID: 0000061403 - 003

Qualification level	Interval [h]	Limit	Item	Maintenance tasks	Option	Task
				Overhaul starter.		W2010
				Fit new seals/sealing materials for all disassembled components.		W2062
				Overhaul charge-air coolant pump.		W2070
				Overhaul engine coolant pump.		W2110
				Check engine coolant thermostat and check thermal actuator, fit new one if necessary.		W2154
QL3	13500	18 a	Fuel delivery pump	Fit new fuel delivery pump.		W1051
QL3	13500	18 a	Battery charging alternator	Overhaul battery-charging generator.		W1043
QL3	13500	18 a	Cylinder heads	Overhaul cylinder heads.		W1134
QL3	13500	18 a	Oil priming pump	Visually check oil priming pump.		W1605
QL3	13500	18 a	Auxiliary PTO	Replace PTO.		W1735
QL3	13500	18 a	Turbochargers	Fit new turbochargers.		W1041
QL4	13500	6 a	Automatic engine oil filter	Remove filter housing and clean. Fit new filter candles.	X	W1035
QL4	13500	18 a	Extended component maintenance	Completely disassemble the engine.Inspect engine components as per assembly instructions and repair or fit new components as required.		W3000
				Replace all elastomeric parts and seals with new ones.		W3001
				Fit new piston rings.		W3002
				Fit new conrod bearings.		W3003
				Fit new crankshaft bearings.		W3004
				Fit new cylinder liners.		W3005
				Fit new high-pressure fuel pump.		W3007
				Fit new camshaft bearings and camshaft thrust bearings.		W3018
				Fit new pressure relief valve in high-pressure fuel system.		W3021
				Replace wiring harnesses.		W3083
				Fit new shaft misalignment coupling.	X	W3097
				Fit new rubber elements of engine mounts.	X	W3100
				Replace swing followers and swing-follower shafts.		W3103
				Check vibration damper, fit new one if necessary.		W3108
				Replace diaphragm coupling.	X	W3112
				Replace pulley.		W3143
				Check engine oil pump, replace if necessary.		W3177
				Gear train: Check tooth flanks for wear (visual inspection), replace bearing bushings.		W3182
QL4	13500	18 a	Engine mounts	Fit new Trunnion mount element.		W1124
QL4	13500	18 a	Exhaust pipe bellows	Replace exhaust pipe bellows.		W1331
QL4	13500	18 a	Steel-spring coupling (Geislinger)	Inspect steel-spring coupling, repair if necessary.	X	X5374

w = weeks
m = months
a = years

TIM-ID: 0000061403 - 003

1.5 Maintenance schedule matrix 10,000 h

0 to 10,000 Operating hours

Task			Operating hours [h]																			
	Limit	Daily	500	1,000	1,500	2,000	2,500	3,000	3,500	4,000	4,500	5,000	5,500	6,000	6,500	7,000	7,500	8,000	8,500	9,000	9,500	10,000
Engine																						
W0800	-																					
W1024	1 w																					
W1974	1 a																					
W1008	2 a																					
W0500	-	X																				
W0501	-	X																				
W0503	-	X																				
W0505	-	X																				
W0506	-	X																				
W0525	-	X																				
W1089	-	X																				
W1009	2 a		X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X	X
W1864	6 m			X		X		X		X		X		X		X		X		X		X
W1463	1 a			X		X		X		X		X		X		X		X		X		X
W1001	2 a			X		X		X		X		X		X		X		X		X		X
W1241	2 a			X		X		X		X		X		X		X		X		X		X
W1575	1 a						X					X					X					X
W1207	2 a			X			X					X					X					X
W1046	2 a											X										X
W1011	6 a											X										X
W1026	6 a											X										X
W1526	18 a											X										X
W1006	18 a											X										X
W1713	18 a											X										X
W1689	18 a											X										X
W1610	18 a																					X
W1755	18 a						X					X					X					X
W1975	18 a										X									X		
W1250	5 a											X										X
W1041	18 a											X										X
W1251	6 a																					X
W2000	18 a																					X
W2001	18 a																					X
W2002	18 a																					X
W2004	18 a																					X
W2007	18 a																					X
W2009	18 a																					X
W2010	18 a																					X
W2062	18 a																					X
W2070	18 a																					X
W2110	18 a																					X
W2154	18 a																					X

w = weeks
m = months
a = years

TIM-ID: 0000061404 - 003